Assignment: Convolutional Neural Networks

DSA 8401: Applied Machine Learning

Master in Data Science and Analytics

Strathmore University

@iLabAfrica Centre

1 Goals

This activity focuses on learning how to use a new architecture that has been extremely successful classifying images. The students must learn:

- Implement Convolutional Neural Netwoks
- Train and evaluate models
- Fine tune hyperparameters
- Classify with the model trained
- Saving and restoring models

2 Assignment Description

In this activity we will use a data set with x-ray images to train a convolutional neural network that classifies whether the patient has pneumonia or not. The description of the data set can be found at the following link:

https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

It is requested:

1. Use Keras to load the dataset

```
# Load the modules needed

import numpy as np
import pandas as pd
import tensorflow as tf
import os
from tensorflow.keras.preprocessing.image import ImageDataGenerator,
                                    load_img
import matplotlib.pyplot as plt
%matplotlib inline
```

```
# Define classes and read the fle's names

basepath = "../input/chest-xray-pneumonia/chest_xray"
classes = ["NORMAL", "PNEUMONIA"]

images = {}
for c in classes:
  images[c] = os.listdir(os.path.join(basepath, "train", c))

print("Number of training images in the class NORMAL: {}".format(len(
                                        images["NORMAL"])))
print("Number of training images in the class PNEUMONIA: {}".format(len(
                                        images["PNEUMONIA"])))
```

```
# Show some example  images with the associate category

fig, ax = plt.subplots(nrows=1, ncols=6, figsize=[24, 8])

for l in range(3):
  for i, c in enumerate(classes):
    j = np.random.randint(1000)
    img = load_img(os.path.join(basepath, "train", c, images[c][j]))
    ax[l*2 + i].imshow(img, cmap="gray")
    ax[l*2 + i].set_title("{}".format(c))
    ax[l*2 + i].axis("off")
```

```
  # Define the processes to read the images from the hard disk

  batch_size = 32

  datagen_train = ImageDataGenerator(rescale=1./255, horizontal_flip=True
                                        )

  training_set = datagen_train.flow_from_directory(os.path.join(basepath,
                                        "train"),
                 target_size=(150, 150),
                 color_mode="grayscale",
                 batch_size=batch_size)

  datagen_test = ImageDataGenerator(rescale=1./255)

  test_set = datagen_test.flow_from_directory(os.path.join(basepath, "
                                        test"),
                 target_size=(150, 150),
                 color_mode="grayscale",
                 batch_size=batch_size)
```

2. Summary of the training set
3. Create a model trying different architectures (number of layers, number of neurons, filters, pooling layers, etc.).
4. Train different models and evaluate. Analize which one perform better.
5. Specify the parameters used at the training (Cost function, optimizer, ...)
6. Show examples of images where the CNN classified wrong.
7. Hypothesis of what made the CNN classify wrongly those images

8. Results and conclusions

3 Hand in

The students will upload in the eLearning platform the Jupyter notebook. The notebook will contain the python code, the plots and especially personal responses to the previous questions. If there is no personal comment the mark will be F. Marks C,B and A will depend on the variety and quality of the personal comments.